MTH1002-B

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

May 2023

Mathematical Methods

Module Leader: Dr Layal Hakim

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No Word Count specified

SECTION A

1. (a) Using Gaussian elimination, find the inverse (if it exists) of the matrix

$$A = \begin{pmatrix} 1 & 0 & -2 \\ 3 & -1 & 4 \\ -2 & 1 & 1 \end{pmatrix}.$$

(8)

(b) Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 1 & -1 \\ -2 & 0 \end{pmatrix}.$$

(8)

(c) Find the critical point of the function

$$f(x, y) = -x^2 - 4xy + 2y^2 + 3x - y + 5,$$

and determine whether it is a local minimum, a local maximum, or a saddle point. (8)

(d) Express the definite integral

$$\int_1^4 (3x + 1)\, dx$$

as a Riemann sum and hence evaluate it. Verify your answer by calculating the integral using the rules of integration. (7)

(e) Evaluate the following definite integrals by using suitable substitutions:

(i)

$$\int_0^{\sqrt{3}} \frac{x}{\sqrt{1 + x^2}}\, dx,$$

(ii)

$$\int_0^4 x\sqrt{16 - x^2}\, dx.$$

(9)

(f) Solve the following ordinary differential equations:

(i)

$$\frac{dy}{dx} - \frac{1}{x}y = 1 + \frac{1}{x^2} \qquad \text{with} \qquad y(1) = \frac{1}{2},$$

(ii)

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = \cos x \qquad \text{with} \qquad y(0) = 1, \ \frac{dy}{dx}(0) = 0.$$

(10)

[**50**]

SECTION B

2. (a) Using elementary row operations, or otherwise, evaluate the determinants of the following matrices:

(i)
$$\begin{pmatrix} 1 & -2 & 0 & 4 \\ -2 & 3 & 1 & 1 \\ 5 & -1 & 2 & -3 \\ 3 & -4 & 1 & -2 \end{pmatrix};$$

(ii)
$$\begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}$$

with constants $a < b < c$. For full marks in part (ii), you should give the determinant as a product of three factors. (13)

(b) Use elementary row operations to put the following system of linear equations into reduced row echelon form:

$$\begin{aligned} -x + 2y - z + 4u &= 1 \\ 3x - y + 5z - u &= -2 \\ 2x + 3y + 6z - 2u &= 0 \end{aligned}$$

Hence, find all solutions to the system (if any exist). If there are infinitely many solutions, you should give the general solution, indicating clearly which values may be chosen arbitrarily. (12)

[**25**]

3. (a) Find the Taylor series expansion of the function

$$f(x, y, z) = e^{x+z} \cos(x + y)$$

about the point $(0, \pi/2, 0)$ in powers of x, $(y - \pi/2)$ and z up to and including the quadratic terms. (11)

(b) Find all of the critical points of the function

$$f(x, y) = [(x^2 - 1)^2 + (y - 1)^2][1 + (y + 1)^2],$$

and use the second derivative test to determine for each of them if it is a local maximum, a local minimum, or a saddle point. If the second derivative test is inconclusive try to classify the critical point otherwise. (14)

[**25**]

4. (a) Calculate the area of the domain $D \in \mathbb{R}^2$ bounded by the curves

$$y = x$$

and

$$y = 2 - x^2.$$

(7)

(b) Evaluate the integral

$$\int_D |xy| \, dxdy$$

over the domain

$$D = \left\{ (x, y) \in \mathbb{R}^2 \;\middle|\; \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \right\}$$

with constants $a > 0$ and $b > 0$. (10)

(c) Suppose that $y = \cos 2x$ is a solution of a third-order linear homogeneous differential equation with constant real coefficients. Determine the coefficients of the differential equation and give its general solution.

Hint: The differential equation is not uniquely determined by the given information; it is a family of differential equations with one free parameter. (8)

[**25**]